Exp-03 3. Load Dataset from csv, Pdf, excel into Sqlite using Python

Aim:-

To load a dataset from csv, excel & pdf files into Mysql database and display the data.

Procedure:-

* Import the required python libraries
* Connect to the Mysql database using SQL Alchemy
* Load dataset from csv, excel and pdf.
* Store each dataset into separate Mysql tables.
* Retrieve and display the stored table contents
* Perform simple web Scrap to count HTML tables from webpage

Libraries Used:-

* Sqlite3
* Pandas
* pdfplumber
* requests
* beautiful Soap

Code:-

```
import sqlite3
import pandas as pd
import pdfplumber
conn = sqlite3.connect("Heart_database.db")
cursor = conn.cursor()
df_csv = pd.read_csv(r"C:\Users\Thesi\Downloads\heart.csv")
df_csv.to_sql("heart_from_csv", conn, if_exists="replace", index=False)
print("Loaded heart_from_csv", len(df_csv), "rows")
df_excel = pd.read_excel(r"C:\Users\Downloads\heart.xlsx")
df_excel.to_sql("heart_from_excel", conn, if_exists="replace",
    index=False)
```

```
print("Loaded heart_from_excel:", len(df_excel), "rows")
columns=["age", "sex","chest", "rest_bp", "cholesterol", "blood_sug
"resting_electrocardio graphic_results", "max_heart_rate_achieved",
"exercise_induced_angina", "oldpeak","slope", no:ofmajor_vessels",
"thal","class"]

all_rows = []
with pdplumber.open(r"C:\Users\Downloads\heart dataset.pdf") as pdf:
    pages = pdf.pages
    for i in range(0, len(pages), 2):
        left = pages[i].extract_tables()[0]
        right = pages[i+1].extract_tables()[0]
        for l_row, r_row in zip(left[1:], right[1:]):
            all_rows.append(l_row + r_row)

df_pdf = pd.DataFrame(all_rows, columns=columns)
df_pdf.to_sql("heart_from_pdf", conn, id_exists="replace", index=F
print("Loaded heart_from_pdf:", len(df_pdf), "rows")
conn.commit()
for table in ["heart_from_csv", "heart_from_excel", "heart_f
    _pdf"]:
        print(f"\n--{table}--")
        print(pd.read_sql(f"SELECT * FROM {table} LIMIT 5"
conn.close()
```

~~import~~
import.

Output: Loaded heart_from_csv: 265 rows
Loaded heart_from_excel: 265 rows
Loaded heart_from_pdf: 265 rows

Age	Sex	Chest	bp	cholestrol	blood_sugar	Cardio_result	max heart rate
70	1	4	130	322	0	2	109
67	0	3	115	564	0	2	160
57	1	2	124	261	0	0	141
64	1	4	128	263	0	0	105
74	0	2	120	269	0	2	121

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0	2.4	2	3	3	present
0	1.6	2	0	7	absent
0	0.3	1	0	7	present
1	0.2	2	1	7	absent
1	0.2	1	1	3	absent

Code_2:-

```
import requests
from bs4 import BeautifulSoup
url = "https://www.w3schools.com/html/html_tables.asp"
response = requests.get(url)
print(response.status_code)
soup = BeautifulSoup(response.text, "html.parser")
tables = soup.find_all("table")
print("Total tables:", len(tables))
```

Output: 200 Total Tables: 2.

Result:
loaded a dataset as csv, excel and pdf into sqlite3 as database and displayed the data successfully.